# Tutorial Sheet IX

**Q.1.** Apply Gram-Schmidt process to the vectors $(1,0,1)$, $(1,0,-1)$ and $(0,3,4)$ to obtain an ONB for $\mathbb{R}^3$.

**Q2.** Consider the vector space $M_{n\times n}(\mathbb{C})$. Show that the map $\langle \cdot , \cdot \rangle : M_{n\times n}(\mathbb{C}) \times M_{n\times n}(\mathbb{C}) \to \mathbb{C}$ defined by
$$\langle A, B \rangle = tr(AB^*)$$
is an inner product on $M_{n\times n}(\mathbb{C})$. Let $Y$ be the space of all diagonal matrices.

(a) Find an ONB for $Y$, and find $Y^\perp$.

(b) For real $n\times n$ matrices $A$ and $B$, show that
$$tr(AB^T)^2 \leq tr(AA^T)\, tr(BB^T).$$

(c) Consider the linear functional defined by
$$\varphi : M_{n\times n}(\mathbb{C}) \to \mathbb{C}$$
$$A = (a_{ij}) \mapsto \sum_{i,j=1}^{n} a_{ij}.$$
Check the validity of the Riesz representation theorem, that is, find the unique matrix $B$ such that
$$\varphi(A) = \langle A, B \rangle, \ \forall A \in M_{n\times n}(\mathbb{C}).$$

(d) Let $J$ be the $n\times n$ matrix with all the entries equal to $1$. Find the best approximation of $J$ in $Y$.

Q3. Let $U$ be a unitary map on $\mathbb{R}^2$, with the standard inner product. Show that the matrix representation of $U$ with respect to the standard basis is

either $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ or $\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$

(rotation by $\theta$) (reflection followed by rotation)

for some $0 \leq \theta \leq 2\pi$.

Q4. For the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

find a orthogonal matrix $U$ s.t. $U^{t}AU = D$, where $D$ is a diagonal matrix.

Q5. Show that $T$ is normal if and only if $T = T_1 + iT_2$ for some Commuting self-adjoint maps $T_1$ and $T_2$.